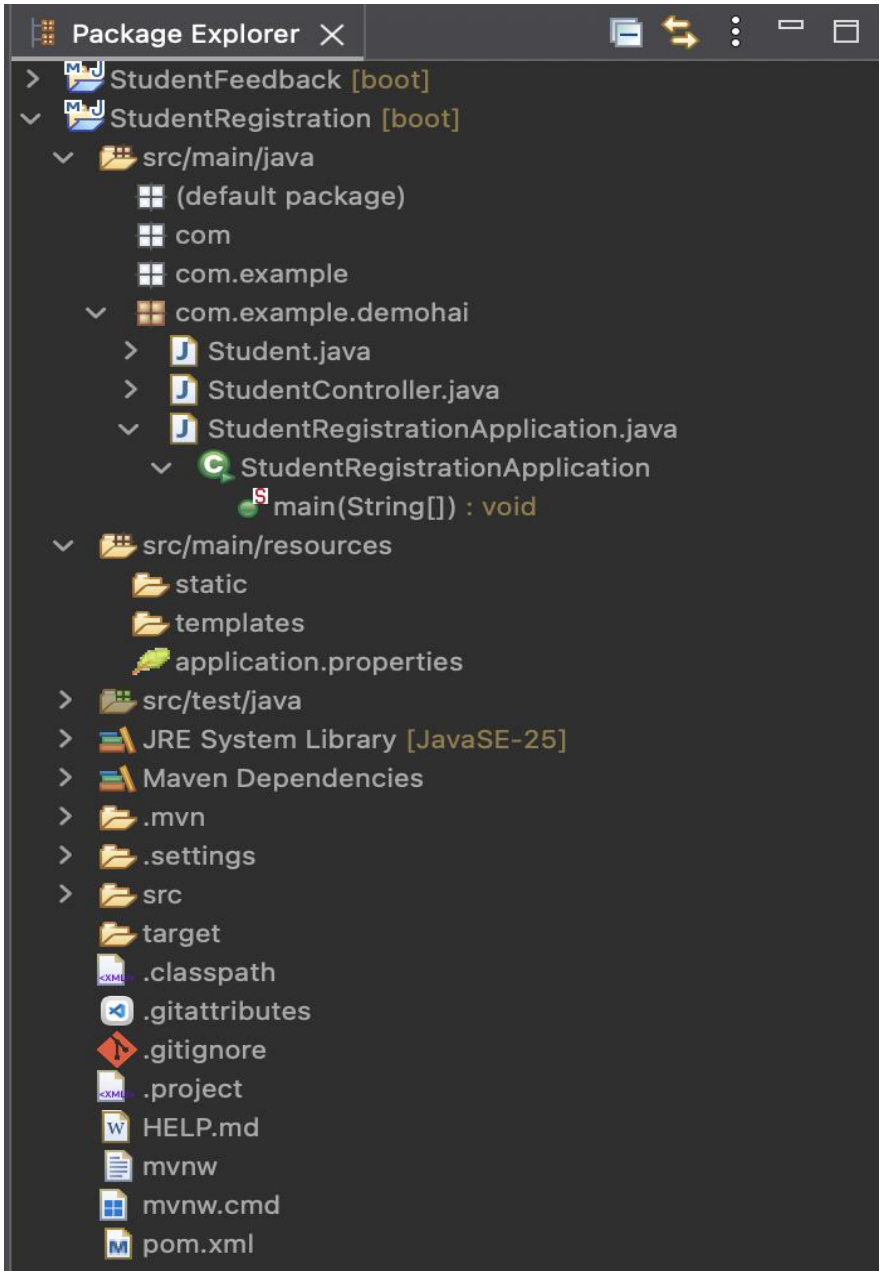
Java SB Case Study 2

~Student Registration System~

1 — Project Structure--



2 — Student.java

StudentRegistrationApplication.java | Student.java | StudentController.java | application.properties

```java

package com.example.demohai;

public class Student {
    private String name;
    private int rollNumber;
    private String course;

    public Student() {
    }

    public Student(String name, int rollNumber, String course) {
        this.name = name;
        this.rollNumber = rollNumber;
        this.course = course;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public int getRollNumber() {
        return rollNumber;
    }

    public void setRollNumber(int rollNumber) {
        this.rollNumber = rollNumber;
    }

    public String getCourse() {
        return course;
    }

    public void setCourse(String course) {
        this.course = course;
    }
}

```

3 — StudentController.java

StudentRegistrationApplication.java | Student.java | StudentController.java | application.properties

```java
package com.example.demohai;

import org.springframework.web.bind.annotation.*;
import java.util.ArrayList;
import java.util.List;

@RestController
@RequestMapping("/students")
public class StudentController {

    private List<Student> students = new ArrayList<>();

    @PostMapping
    public String addStudent(@RequestBody Student student) {
        students.add(student);
        return "Student registered successfully";
    }

    @GetMapping
    public List<Student> getAllStudents() {
        return students;
    }

    @PutMapping("/{rollNumber}")
    public String updateStudent(@PathVariable int rollNumber,
                                @RequestBody Student updatedStudent) {
        for (Student s : students) {
            if (s.getRollNumber() == rollNumber) {
                s.setName(updatedStudent.getName());
                s.setCourse(updatedStudent.getCourse());
                s.setRollNumber(updatedStudent.getRollNumber());
                return "Student updated successfully";
            }
        }
        return "Student not found";
    }

    @DeleteMapping("/{rollNumber}")
    public String deleteStudent(@PathVariable int rollNumber) {
        boolean removed = students.removeIf(
            s -> s.getRollNumber() == rollNumber
        );

        if (removed) {
            return "Student deleted successfully";
        }

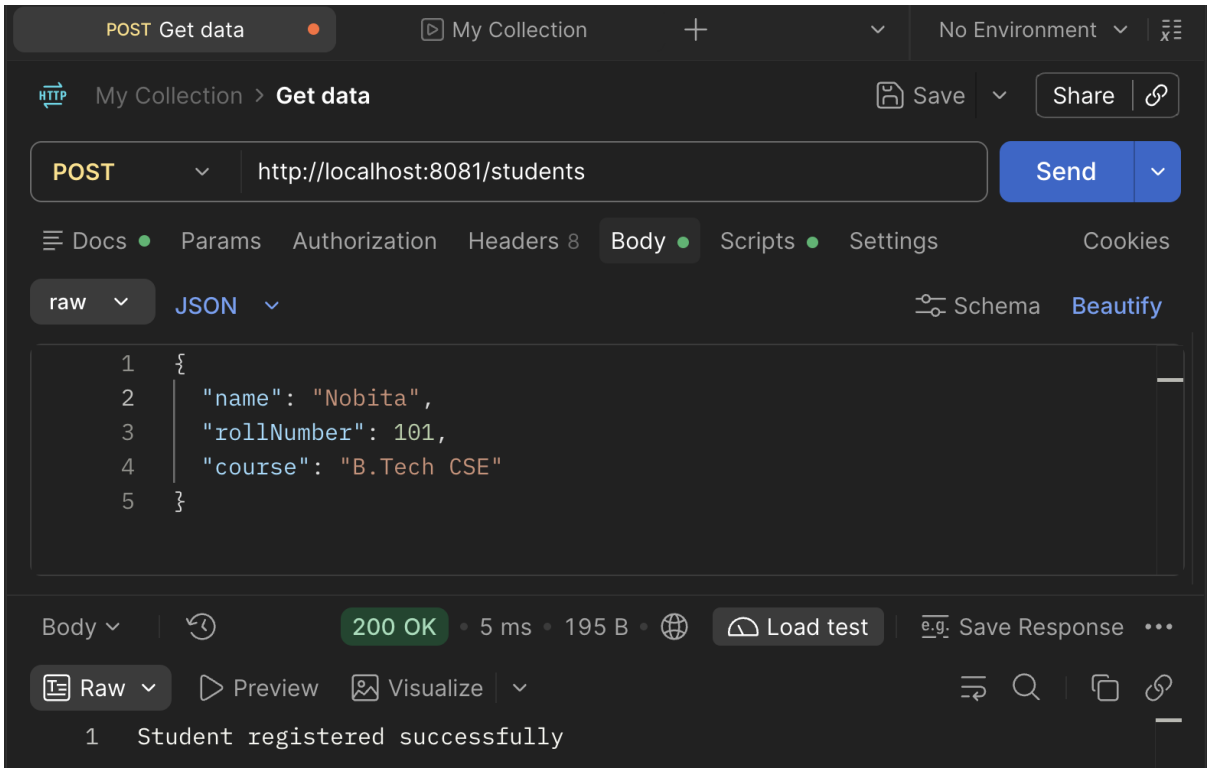
        return "Student not found";
    }
}

```

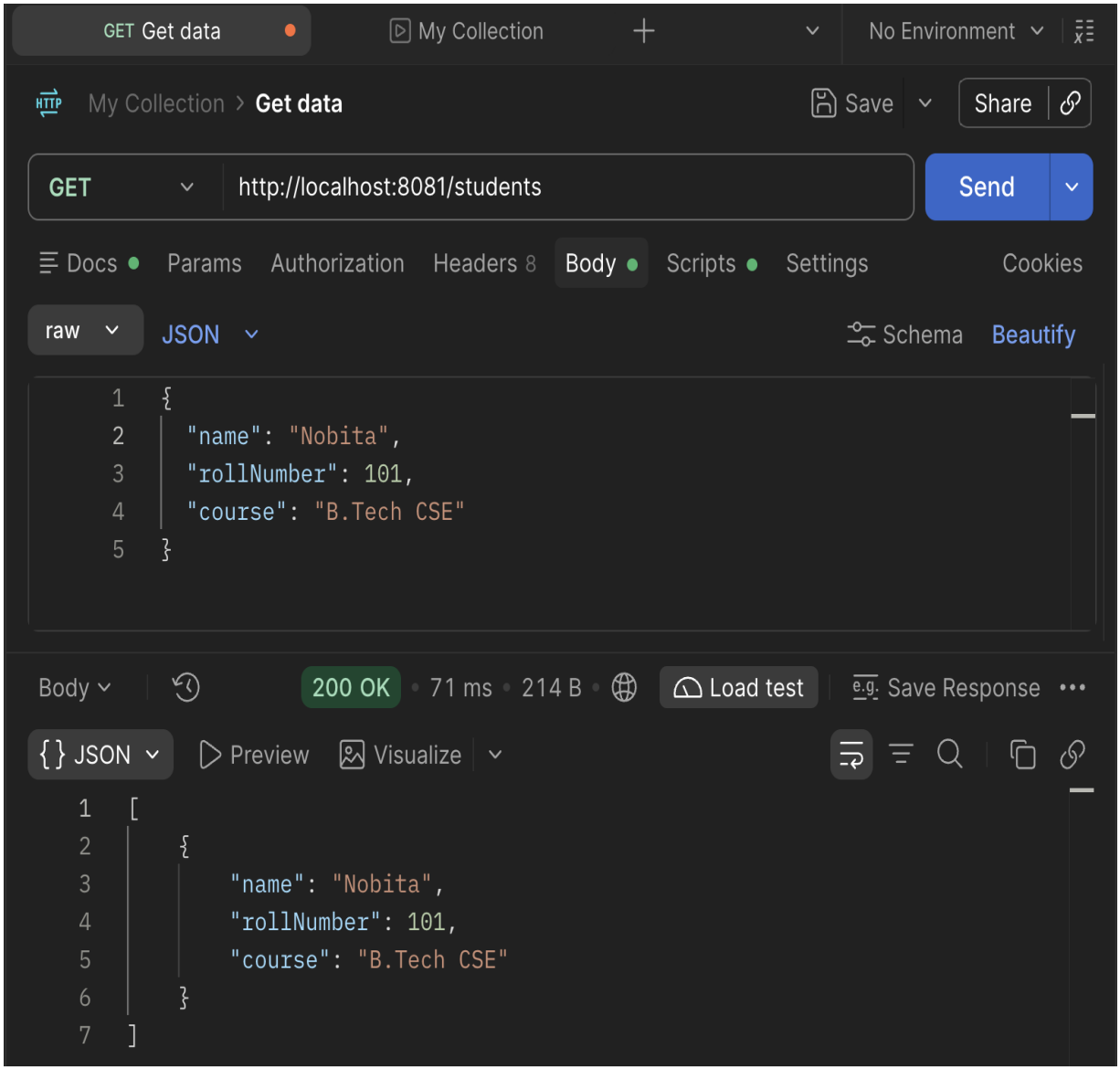
4 — application. Properties



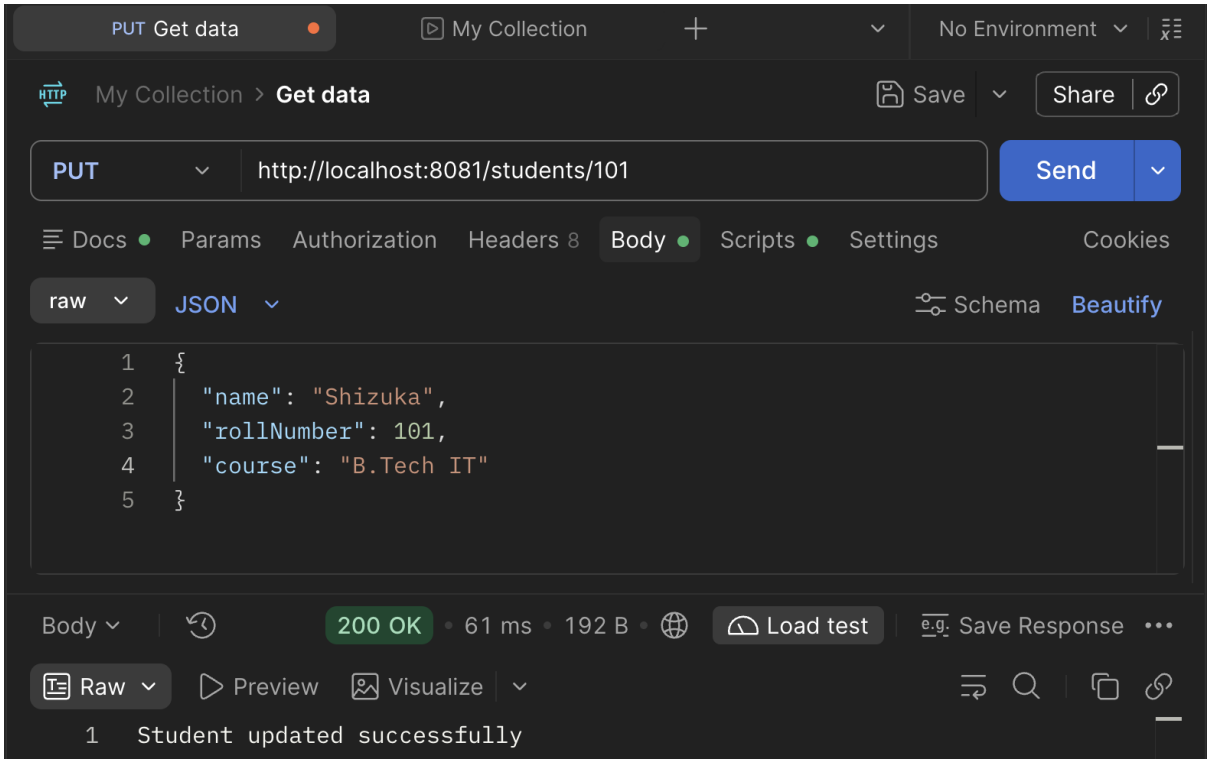
5 — POST (Register Student)



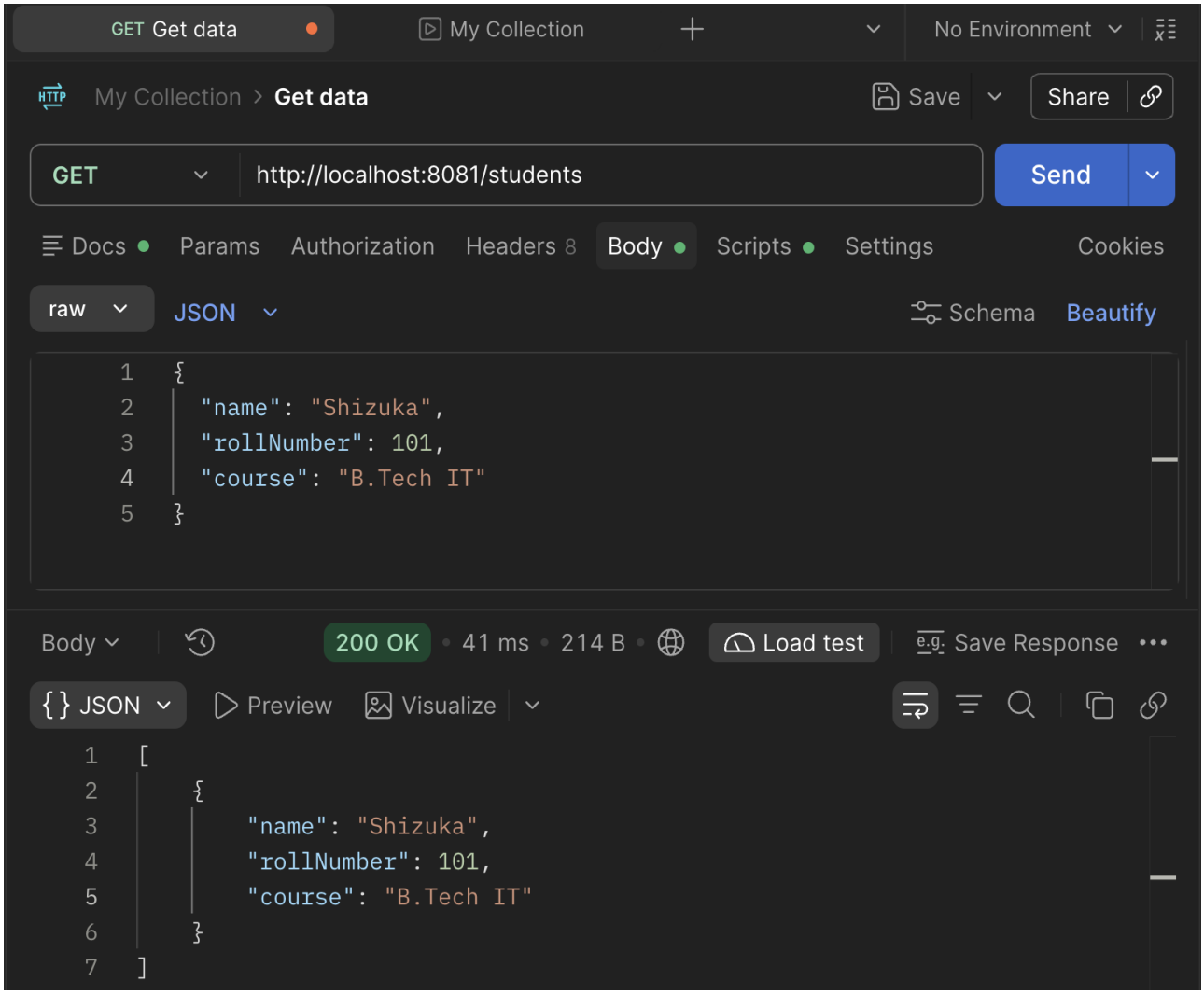
6 — GET (Display All Students)



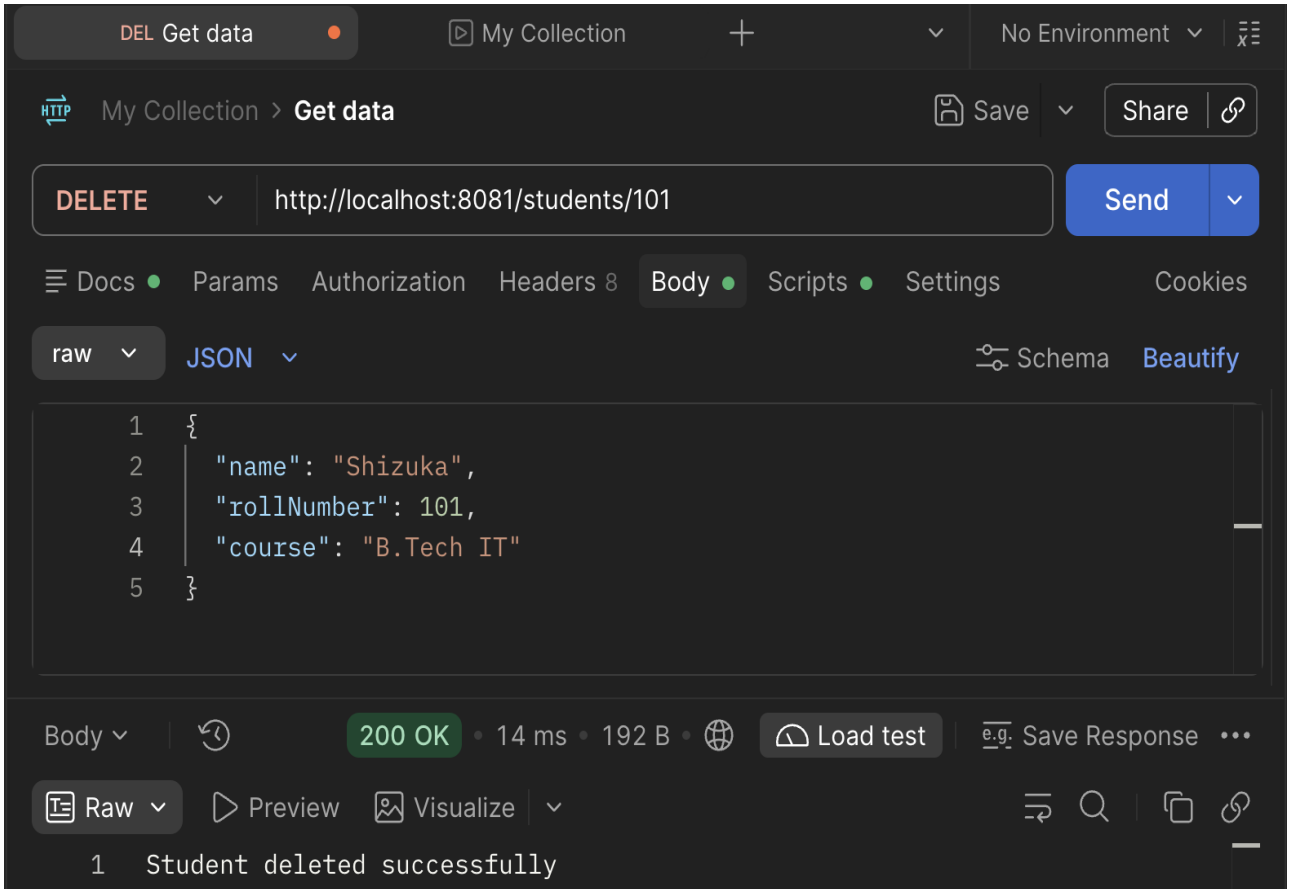
7 — PUT (Update Student)



~// After Updating--



8 — DELETE (Delete Student)



~// After Deleting--

